POViewer

Point of View Viewer

Design Specification — v1.0 — BKC AI Triad Project — 2026

POV Color Convention

🟢 Accelerationist	🔴 Safetyist	🟡 Skeptic	🟣 Cross-Cutting

⬤ Multi-POV (badge)	✅ Agrees ⚠️ Contradicts

1. Product Overview

POViewer is a desktop research tool that lets analysts load source documents — web URLs and PDFs — and instantly see how the claims within those documents map to a set of configurable Point of View (POV) taxonomies. Each taxonomy defines a camp's Goals/Values, Methods, and Data/Facts. Highlights in the document viewer show which claims land where across those camps, and a dedicated POV panel lets analysts filter, explore, and drill into the mappings in detail.

The tool is designed first for the BKC AI Triad research team and will later be released publicly. The MVP is a static prototype with hardcoded data — no live AI calls — built to prove the three-pane interaction model before connecting real AI analysis.

Attribute	Decision
Platform	Desktop app — Mac and Windows (Electron)
Primary users (v1)	Internal BKC research team
Public release	Phase 2 — after internal validation
AI provider	User supplies own API key (Gemini or Claude)
MVP scope	Static prototype — hardcoded data, no live AI
Taxonomy model	Taxonomy-agnostic: user loads any POV JSON files
Session persistence	Taxonomy file paths and Notebook state remembered across sessions

2. Core Concepts & Terminology

2.1 Notebook

A Notebook is the top-level organizing unit, equivalent to a project or investigation. Each Notebook has its own source list, its own loaded taxonomy set, and its own analysis results. Notebooks are independent of each other. Multiple Notebooks can exist; one is active at a time.

Notebook contains	Description
Name + description	User-defined label and optional notes
Source list	All documents added to this Notebook (URLs and PDFs)
Active taxonomy set	Which POV JSON files are loaded and enabled
Analysis cache	Stored AI analysis results per source, keyed by taxonomy version
View state	Which sources are selected, which POVs are toggled on

2.2 Source

A Source is any document added to a Notebook. Sources can be web URLs or PDF files. A source has two states: deselected (listed but not analyzed or rendered) and selected (rendered in Pane 2 and analyzed against the active taxonomies).

2.3 Point

A Point is the atomic unit of analysis. The AI segments each source document into logical claims — units of meaning that represent a distinct assertion, argument, or piece of evidence. Points are not fixed to sentence or paragraph boundaries; the AI decides where one point ends and the next begins based on semantic coherence.

> **Why AI-decided segmentation?**
> Sentence boundaries are too granular — a single claim often spans multiple sentences. Paragraph boundaries are too coarse — a paragraph may contain three distinct arguments. AI-decided segmentation finds the natural claim boundaries, which is what the taxonomy mapping needs to be meaningful.

2.4 Mapping

A Mapping is the relationship between one Point and one taxonomy node. A single Point can have multiple Mappings — to nodes in different POV camps, or to multiple nodes within the same camp. Each Mapping has two attributes:

- Alignment: does this Point AGREE with or CONTRADICT the taxonomy node?
- Strength: how strongly does the Point map to this node? (HIGH / MEDIUM / LOW)

2.5 Taxonomy

A Taxonomy is a JSON file that defines one POV camp's nodes across three categories: Goals/Values, Methods, and Data/Facts. POViewer is taxonomy-agnostic — any conforming JSON file can be loaded. The AI Triad taxonomy (four files: accelerationist, safetyist, skeptic, cross-cutting) is the default set, but users can substitute or supplement with any domain.

3. Three-Pane Layout

The application window is divided into three resizable panes arranged horizontally. Each pane has a distinct role. The panes are always visible simultaneously — there is no tab-switching between them.

PANE 1 **Sources** ~23% of window	PANE 2 **Source Viewer** ~44% of window	PANE 3 **POV Panel** ~33% of window
Notebook switcher Source list (add / remove) Select / deselect each source Analysis status badge Taxonomy manager	Rendered document text Highlighted points with POV chip + agree/contradict icon Click highlight → activates Point in Pane 3 Scrolls independently	POV filter toggles (top) Detail card for selected Point Per-camp mapping breakdown Taxonomy node links

4. Pane 1 — Sources

4.1 Notebook Bar

A horizontal bar at the very top of Pane 1 shows the name of the active Notebook with a dropdown to switch between Notebooks or create a new one. A settings icon opens the Notebook settings (name, description, taxonomy assignments).

4.2 Source List

Below the Notebook bar is a scrollable list of all sources in the active Notebook. Each source row contains:

Element	Description
Checkbox	Selects/deselects the source for rendering and analysis
Favicon / type icon	URL sources show site favicon; PDFs show a PDF icon
Title	Auto-extracted page title or filename. Editable on double-click.
Status badge	One of: Not analyzed • Analyzing... • Analyzed • Stale (taxonomy changed)
Overflow menu (⋮)	Options: Re-analyze, Open original URL, Remove from Notebook, Copy citation

4.3 Add Source

A prominent '+ Add Source' button at the bottom of the source list opens a modal with two tabs:

- URL tab: paste any web URL. On confirm, POViewer fetches the page, extracts readable text, saves a Markdown snapshot, and adds the source.
- PDF tab: file picker dialog. On confirm, the PDF is copied into the Notebook's local storage folder and added.

4.4 Taxonomy Manager

A collapsible 'Taxonomies' section at the bottom of Pane 1 shows the currently loaded POV files. Each row shows the POV name, a colored dot, and an enable/disable toggle. Controls:

- Load taxonomy: file picker for any conforming JSON file. The file path is remembered across sessions.
- Reload: re-reads the file from disk (picks up edits made externally).
- Remove: unloads the taxonomy from this Notebook.
- Color swatch: click to change the color assigned to this POV camp.

Session persistence

> Loaded taxonomy file paths are stored in the app’s local settings file (not per-Notebook). When the app relaunches, it attempts to reload each taxonomy from its remembered path. If a file has moved, a warning badge appears and the user is prompted to re-locate it.

5. Pane 2 — Source Viewer

5.1 Document Rendering

When a source is selected, its Markdown snapshot is rendered as clean readable prose in Pane 2. PDFs are rendered page by page with a continuous scroll. The toolbar at the top of Pane 2 provides: source title, a ‘View original’ link, zoom controls, and a search field.

5.2 Point Highlight System

After AI analysis completes, the document text in Pane 2 is annotated with inline highlights. Each highlight corresponds to one Point — one AI-identified logical claim. The highlight system has been designed for maximum legibility with minimum visual noise.

5.2.1 Single-POV Point

When a Point maps to exactly one POV camp, the highlight uses that camp’s color as a background fill on the text span, plus an agree/contradict icon at the right edge of the highlight.

Text in document	POV chip	Icon	Taxonomy node
Accelerating AI deployment reduces the opportunity cost of unsolved diseases.	ACCEL	✅	*acc-goals-001: Abundance through AI*
We cannot currently inspect the internal goals of a large neural network.	SAFETY	✅	*saf-data-003: Interpretability insufficient*
Current AI hiring tools show 23% higher rejection rates for minority applicants.	SKEPTIC	✅	*skp-data-001: Measurable bias*
Safety-focused labs moving slowly only hands the race to less careful actors.	ACCEL	⚠️	*saf-methods-001: Contradicts pause argument*

5.2.2 Multi-POV Point

When a Point maps to two or more POV camps, the highlight uses a neutral slate color (#5D6D7E) with a numeric badge showing the count of POV camps involved. The agree/contradict icon reflects the dominant alignment. Clicking the highlight opens Pane 3 detail view, which shows the full breakdown.

Text in document	POV chip	Icon	Note
Defining 'harm' is the central challenge for any AI regulatory framework.	3 POVs	⚠️	*Maps to cc-004 (Harm) — all three camps*
AGI timelines determine whether we have years or decades to solve alignment.	2 POVs	✅	*Maps to acc-data-002 + saf-data-002*

5.2.3 Visual Hierarchy Summary

Scenario	Highlight color	Icon	Badge
Single POV, agrees	POV camp color (solid fill)	✅	None
Single POV, contradicts	POV camp color (solid fill)	⚠️	None
Multi-POV (2+ camps)	Neutral slate #5D6D7E	⚠️ or ✅	Number (e.g. '2 POVs')
Not yet analyzed	None	None	None
Currently selected Point	White ring / focus outline	Same	Same

5.3 POV Filter Bar

A slim horizontal filter bar sits just above the document text in Pane 2. It contains one toggle chip per loaded taxonomy, using each camp's color. Toggling a camp OFF removes all highlights for that camp from view — the text is still there, the highlight simply disappears. This allows analysts to visually isolate one camp's footprint across the document.

The filter bar also has an 'All' toggle and a 'None' toggle for quick show/hide of all highlights.

> **Filter state is local to Pane 2**
> Toggling a POV in the Pane 2 filter bar does NOT affect the Pane 3 detail view. Pane 3 always shows the full mapping for the selected Point regardless of which filters are active in Pane 2. This prevents the detail panel from going mysteriously blank when a camp is filtered out.

6. Pane 3 — POV Panel

Pane 3 has two modes that coexist: a persistent filter/summary section at the top, and a dynamic detail card below it that updates when a Point is selected in Pane 2.

6.1 Top Section: POV Filter Toggles

Mirrors the filter bar in Pane 2. Toggling a camp here has the same effect: highlights for that camp disappear in Pane 2. The two filter controls (Pane 2 bar and Pane 3 toggles) are kept in sync — changing one changes the other.

Below the toggles, a mini summary shows counts for the active source:

- Total points identified: N
- Points mapped to each active POV camp (with colored dot)
- Points with contradictions: N
- Unmapped points: N

6.2 Detail Card: Selected Point

When no Point is selected, the detail area shows placeholder text: ‘Click any highlighted passage in the document to see its POV analysis.’

When a Point is clicked in Pane 2, the detail card shows:

6.2.1 Point Header

Element	Content
Point text	The exact text span of the selected Point, quoted in full
Location	Document section / page reference (e.g. ‘Section 3, paragraph 2’)
POV count badge	e.g. ‘3 POV mappings’ — total number of mappings for this Point
Navigation arrows	← Previous point | Next point → (cycles through all Points in document order)

6.2.2 Per-Camp Mapping Blocks

For each POV camp that this Point maps to, a collapsible block appears. The block’s left border is colored with that camp’s color. Inside each block:

Field	Content
Camp header	Camp name + color chip + overall stance for this point (AGREES / CONTRADICTS)
Taxonomy category	Which axis this maps to: Goals/Values, Methods, or Data/Facts

Field	Content
Node ID + label	e.g. ‘acc-goals-001: Abundance through AI’ — hyperlinks to taxonomy node definition
Node description	The taxonomy node’s full description text (collapsed by default, expand on click)
Mapping explanation	One sentence from the AI explaining WHY this Point maps to this node
Strength	HIGH / MEDIUM / LOW — how strongly the AI assessed the mapping

6.2.3 Cross-Cutting Alert

If the selected Point maps to the same concept (same cross-cutting node, or two nodes with the same label in different camps) with different alignments across camps — for example, one camp AGREES and another CONTRADICTS — a highlighted alert box appears between the blocks:

> ⚠️ **Vocabulary Collision Detected**
> This passage uses the term ‘harm’ in a way that aligns with the Safetyist interpretation (existential risk) but contradicts the Skeptic interpretation (present, measurable discrimination). This is a cross-cutting vocabulary conflict — the word is shared, the meaning is not.

6.2.4 Unmapped Points

If the Point has no mappings to any taxonomy node (the AI flagged it as outside the taxonomy), the detail card shows an ‘UNMAPPED’ badge and the AI’s suggested category and POV camp for a potential new node. This feeds directly into the taxonomy expansion workflow.

7. Notebooks

7.1 Creating and Managing Notebooks

Notebooks are created from the Notebook bar dropdown. Each Notebook is stored as a local folder on disk containing:

File / folder	Contents
notebook.json	Name, description, creation date, active taxonomy file paths, view state
sources/	One subfolder per source: metadata.json + snapshot.md + raw file
analysis/	One JSON file per source containing the AI-generated Point mappings
taxonomy-cache/	Snapshots of the taxonomy files as they were when analysis was last run

7.2 Source States

State	Description	Visual indicator
Not added	Not in this Notebook	Not shown
Deselected	In the list but not rendered or analyzed	Gray row, unchecked checkbox
Selected	Rendered in Pane 2, analysis triggered if not yet run	Bold row, checked checkbox, blue border
Analyzing	AI analysis in progress	Spinner badge on source row
Analyzed	Analysis complete, highlights visible	Green dot badge
Stale	Taxonomy changed since last analysis, re-analysis needed	Orange ‘Stale’ badge
Error	Fetch failed or analysis failed	Red error badge + tooltip

7.3 Multi-Source View

Only one source is rendered in Pane 2 at a time. However, multiple sources can be selected simultaneously. When multiple sources are selected, Pane 1 shows a source tab bar at the top of Pane 2 — click any tab to bring that source into the viewer without deselecting the others. This allows fast comparison of how different documents handle the same concepts.

8. AI Analysis Pipeline

This section describes the analysis flow for the production version. The MVP uses hardcoded data in place of real AI calls — the architecture described here is what the hardcoded data simulates.

8.1 API Key Configuration

On first launch, POViewer prompts the user to enter an API key for their preferred AI provider (Gemini or Claude). The key is stored in the OS keychain (not in any Notebook file) and is never written to disk in plain text. The user can switch providers or update keys in Settings.

8.2 Analysis Trigger

Analysis is triggered when a source transitions from deselected to selected AND no current analysis exists for the active taxonomy version. If a stale analysis exists (taxonomy was updated), the user is shown a 'Re-analyze' prompt rather than auto-triggering.

8.3 Analysis Steps

1. Fetch and snapshot the source (URL → Markdown; PDF → text extraction).
2. Segment the document into Points using the AI. Each Point gets a stable ID and a character-offset range in the snapshot.
3. For each Point, call the AI with the full taxonomy context and ask it to: identify all matching nodes, determine alignment (agrees/contradicts), assign strength (HIGH/MEDIUM/LOW), and flag cross-cutting collisions.
4. Write results to analysis/<source-id>.json, keyed by taxonomy version.
5. Render highlights in Pane 2 using the character-offset ranges.

8.4 Analysis Output Schema (per Point)

The JSON stored for each analyzed Point:

Field	Type	Description
point_id	string	Stable ID, e.g. 'p-007'
text	string	The exact text of this Point
char_start	integer	Start offset in snapshot.md
char_end	integer	End offset in snapshot.md
mappings	array	List of Mapping objects (see below)
unmapped	boolean	True if no taxonomy node matched
unmapped_hint	object	Suggested POV + category + reason for new node

Each Mapping object:

Field	Type	Description
node_id	string	Taxonomy node ID, e.g. ‘acc-goals-001’
pov	string	Camp name: accelerationist / safetyist / skeptic / cross-cutting
category	string	Goals/Values | Methods | Data/Facts
alignment	string	agrees | contradicts
strength	string	HIGH | MEDIUM | LOW
explanation	string	One sentence: why this mapping was made

9. MVP: Static Prototype Specification

The MVP proves the three-pane interaction model with zero live AI calls. All analysis data is hardcoded. The goal is to demonstrate the complete UX — highlighting, clicking, Pane 3 detail view, POV filtering, taxonomy loading — so the team can validate the design before investing in the AI pipeline.

9.1 Hardcoded Content

Item	MVP approach
Source document	One hardcoded source: a snapshot of a real AI policy article (text pasted in code)
Points	15–20 hardcoded Points with character offsets mapped to the source text
Mappings	Manually authored mappings covering all four camps, agrees and contradicts, single and multi-POV
Taxonomy	AI Triad taxonomy JSON files loaded from disk (real files — taxonomy loading IS wired up)
Analysis trigger	Simulated: clicking a source shows a fake 'Analyzing' spinner for 1.5s then renders highlights
API key	Not required — settings screen exists but is non-functional in MVP

9.2 MVP Feature Checklist

Feature	In MVP?	Notes
Three-pane layout, resizable	✓ Yes	Core layout must work
Notebook create / switch	✓ Yes	At least 2 hardcoded Notebooks to demo switching
Add source (URL + PDF)	✗ No	Source list is hardcoded
Source select / deselect	✓ Yes	Checkbox toggles highlighting on/off
Document text rendered in Pane 2	✓ Yes	Hardcoded snapshot text
Point highlights with color + icon	✓ Yes	Core visual to validate
Multi-POV badge on highlight	✓ Yes	At least 3 multi-POV Points hardcoded
Click highlight → Pane 3 detail	✓ Yes	Core interaction to validate
POV filter toggles (Pane 2 + Pane 3)	✓ Yes	Show/hide highlighting per camp
Pane 3 per-camp mapping blocks	✓ Yes	With agrees / contradicts styling
Cross-cutting collision alert	✓ Yes	At least 1 hardcoded collision case
Unmapped Point display	✓ Yes	At least 1 hardcoded unmapped Point

Feature	In MVP?	Notes
Taxonomy file loading from disk	✓ Yes	Real JSON loading — validates the data model
Taxonomy session persistence	✓ Yes	File paths remembered on relaunch
Live AI analysis	✗ No	Phase 2
PDF rendering	✗ No	Phase 2
Export / sharing	✗ No	Phase 2

9.3 Recommended Technology Stack

Layer	Recommendation	Reason
Shell	Electron	Cross-platform Mac + Windows from one codebase
UI framework	React + TypeScript	Component model fits three-pane layout; strong ecosystem
Styling	Tailwind CSS	Rapid layout iteration; easy to theme POV colors
State	Zustand	Lightweight; no Redux boilerplate for prototype scale
File I/O	Electron main process + Node.js fs	Direct disk access for Notebook folders and taxonomy files
PDF text	pdfjs-dist (Phase 2)	Mozilla's PDF.js — best-in-class browser/Electron PDF rendering

10. Design Principles

Principle	What it means for POViewer
The document is primary	Pane 2 gets the most horizontal space. Highlights are designed to annotate the reading experience, not interrupt it. The filter bar lets users turn all annotations off to read cleanly.
Complexity lives in Pane 3, not Pane 2	The agree/contradict icon in Pane 2 is the most complexity a highlight carries. All the nuance — which node, which camp, why, how strong — lives in the Pane 3 detail card. Pane 2 is scannable; Pane 3 is analytical.
Multi-POV = neutral, not noisy	When a Point touches multiple camps, the slate/neutral color signals 'this is interesting' without pre-judging which camp is dominant. The badge count gives enough information to decide whether to click in.
Vocabulary collisions are first-class	Cross-cutting collision alerts in Pane 3 are not error states — they are the most analytically valuable finding the tool can surface. They should feel like discoveries, not warnings.
Taxonomy-agnostic by design	No assumption about the AI Triad taxonomy is baked into the application code. Colors, camp names, node IDs, and category labels all come from the loaded JSON files. A user could load a political left/right taxonomy or a climate policy taxonomy and the tool would work identically.
Silence is surfaced	The Pane 3 summary section shows not just what is mapped but what is NOT mapped — which camps have zero Points in this document. A document that never touches the Skeptic camp is making an implicit choice; POViewer makes that visible.

11. Open Questions for Phase 2

Question	Why it matters
Multi-source aggregation view: can Pane 3 show mappings ACROSS sources, not just within one?	Enables the ‘how do 10 documents treat this taxonomy node’ query — core to the research use case
Can a user manually override an AI mapping — add, remove, or correct a highlight?	Essential for research trust; analysts need to be able to say ‘the AI got this wrong’
Should Notebooks be shareable / exportable as a bundle?	Required for BKC collaborators and workshop participants to share annotated document sets
How does the tool handle very long documents (100+ pages)?	Analysis cost + rendering performance — chunking strategy and progressive rendering needed
Should Pane 3 include a ‘Taxonomy Gap’ view listing all UNMAPPED points across all sources?	Would make the taxonomy expansion workflow first-class rather than buried in individual Point cards
Web-hosted version vs. desktop only for public release?	Desktop requires install friction; web requires a backend. Decision affects Phase 2 architecture entirely